

Arrow routing

make-arrow and draw-arrow

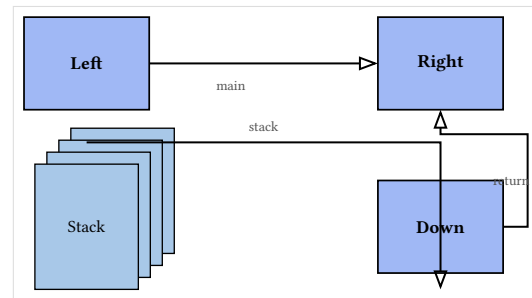
Code

```
#graph-canvas({  
  let left = make-box("Left", pos: (1.5,  
0.8))  
  let right = make-box("Right", pos:  
(8.0, 0.8))  
  let down = make-box("Down", pos: (8.0,  
-2.2))  
  let stack = make-dataset("Stack",  
images: 4, pos: (1.5, -2.2))
```

```
draw-node(left)  
draw-node(right)  
draw-node(down)  
draw-node(stack)
```

```
let a = make-arrow(  
  left,  
  right,  
  out-side: "right",  
  in-side: "left",  
  auto-spacing: true,  
  spacing: 0.6,  
  mode: "hvh",  
  mode-shift: (2, 1.4),  
  label: [main],  
  label-side: "below",  
  label-gap: 0.35,  
  label-dx: 0.15,  
  label-dy: -0.05,  
)  
let b = make-arrow(  
  stack,  
  down,  
  out-side: "top",  
  in-side: "bottom",  
  from-outer: true,  
  from-outer-value: 0.4,  
  to-outer: true,  
  to-outer-value: 0.3,  
  auto-spacing: false,  
  mode: "hv",  
  mode-shifts: (1.2, 0.8),  
  label: [stack],  
)  
let c = make-arrow(  
  down,  
  right,  
  out-side: "right",  
  in-side: "bottom",  
  mode: "vh",  
  label: [return],  
)
```

Output



```

    draw-arrow(a)
    draw-arrow(b)
    draw-arrow(c)
  })

```

edge-label

Code

```

#graph-canvas({
  draw.line((1, 0), (6, 0), mark: (end:
">"), stroke: (paint: black, thickness:
0.75pt))
  edge-label((1, 0), (6, 0), [input],
side: "above", gap: 0.30, dx: 0.2, dy:
0.0, size: 0.5em)
})

```

Output



draw-arrows side distribution

Code

```

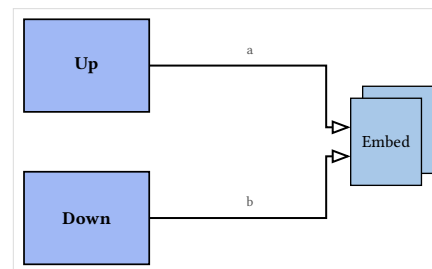
#graph-canvas({
  let up = make-box("Up", pos: (1.5,
1.4))
  let down = make-box("Down", pos: (1.5,
-1.4))
  let target = make-dataset("Embed",
images: 2, image-size: (1.3, 1.6), pos:
(7.0, 0.0))

  draw-node(up)
  draw-node(down)
  draw-node(target)

  let a = make-arrow(up, target, out-
side: "right", in-side: "left", label:
[a])
  let b = make-arrow(down, target, out-
side: "right", in-side: "left", label:
[b])
  draw-arrows((a, b))
})

```

Output



draw-arrows side distribution (4 arrows)

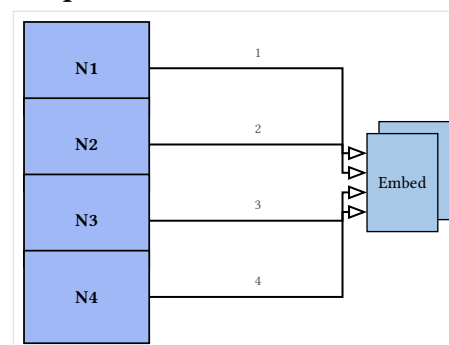
Code

```

#graph-canvas({
  let n1 = make-box("N1", pos: (1.2,
2.1))
  let n2 = make-box("N2", pos: (1.2,
0.7))
  let n3 = make-box("N3", pos: (1.2,
-0.7))
  let n4 = make-box("N4", pos: (1.2,
-2.1))
  let target = make-dataset("Embed",
images: 2, image-size: (1.3, 1.8), pos:
(7.0, 0.0))

```

Output



```

draw-node(n1)
draw-node(n2)
draw-node(n3)
draw-node(n4)
draw-node(target)

let arrows = (
  make-arrow(n1, target, out-side:
"right", in-side: "left", label: [1]),
  make-arrow(n2, target, out-side:
"right", in-side: "left", label: [2]),
  make-arrow(n3, target, out-side:
"right", in-side: "left", label: [3]),
  make-arrow(n4, target, out-side:
"right", in-side: "left", label: [4]),
)
draw-arrows(arrows)
})

```

draw-arrows mixed in/out side distribution

Code

```

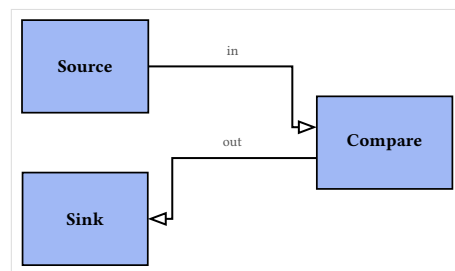
#graph-canvas({
  let source = make-box("Source", pos:
(1.5, 1.4))
  let sink = make-box("Sink", pos: (1.5,
-1.4))
  let compare = make-box("Compare", pos:
(7.0, 0.0))

  draw-node(source)
  draw-node(sink)
  draw-node(compare)

  let arrows = (
    make-arrow(source, compare, out-side:
"right", in-side: "left", label: [in]),
    make-arrow(compare, sink, out-side:
"left", in-side: "right", label: [out]),
  )
  draw-arrows(arrows)
})

```

Output



draw-arrows explicit endpoint order

Code

```

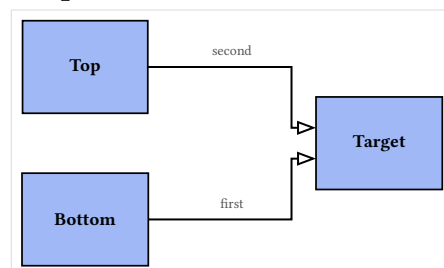
#graph-canvas({
  let top = make-box("Top", pos: (1.6,
1.4))
  let bottom = make-box("Bottom", pos:
(1.6, -1.4))
  let target = make-box("Target", pos:
(7.0, 0.0))

  draw-nodes((top, bottom, target))

  let arrows = (
    make-arrow(top, target, in-side:

```

Output



```

"left", order: 2, label: [second]),
  make-arrow(bottom, target, in-side:
"left", order: 1, label: [first]),
)
draw-arrows(arrows)
})

```

draw-arrows same order uses code order

Code

```

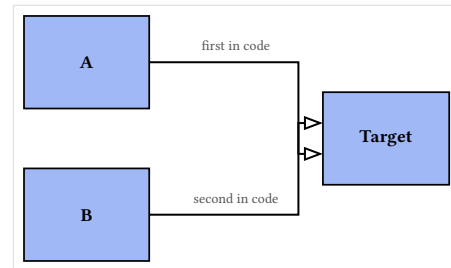
#graph-canvas({
  let a = make-box("A", pos: (1.5, 1.4))
  let b = make-box("B", pos: (1.5, -1.4))
  let target = make-box("Target", pos:
(7.0, 0.0))

  draw-nodes((a, b, target))

  let arrows = (
    make-arrow(a, target, in-side:
"left", order: 1, label: [first in
code]),
    make-arrow(b, target, in-side:
"left", order: 1, label: [second in
code]),
  )
  draw-arrows(arrows)
})

```

Output



draw-arrows with auto-distribute false

Code

```

#graph-canvas({
  let up = make-box("Up", pos: (1.6,
1.2))
  let down = make-box("Down", pos: (1.6,
-1.2))
  let target = make-dataset("Embed",
images: 2, image-size: (1.3, 1.6), pos:
(7.0, 0.0))

  draw-node(up)
  draw-node(down)
  draw-node(target)

  let arrows = (
    make-arrow(up, target, out-side:
"right", in-side: "left", label: [a]),
    make-arrow(down, target, out-side:
"right", in-side: "left", label: [b]),
  )
  draw-arrows(arrows, auto-distribute:
false)
})

```

Output

